Université: Institut Universitaire des Sciences (IUS)

TD N^0 6: Réseau1

Nom &Prénom: SAINT-JEAN Marc-Evenort

Professeur: Ismael SAINT AMOUR

Niveau: 3ème Année

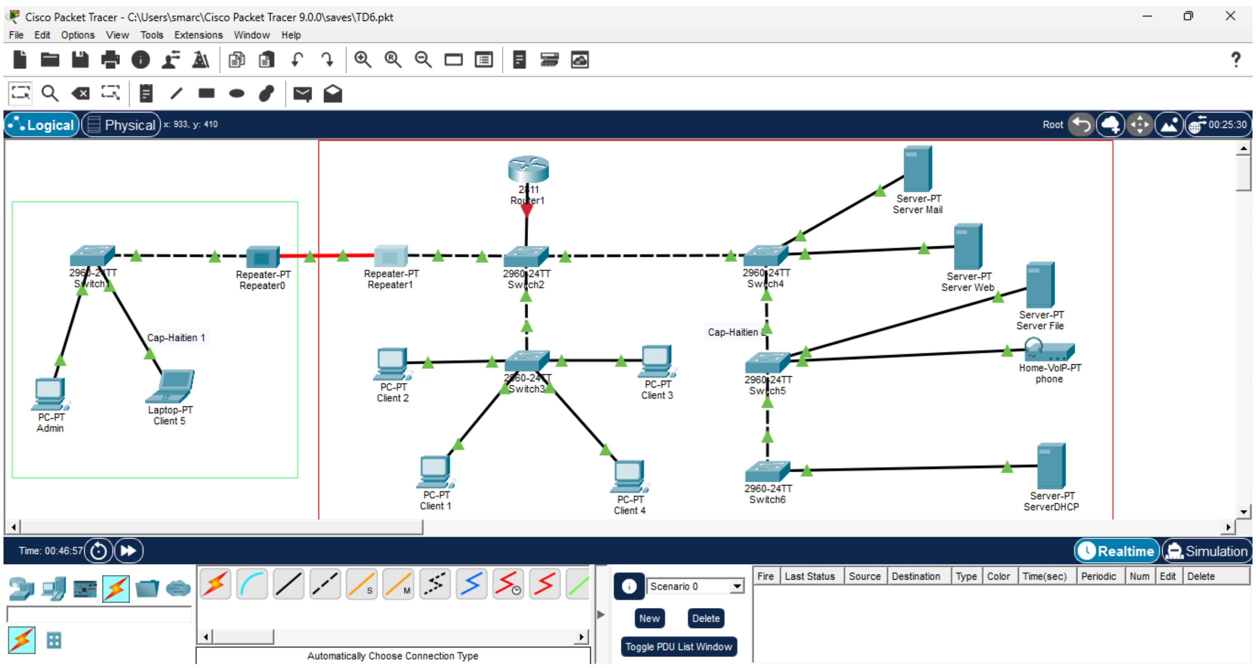
Date: Le 08 /Jan/2026

Objectif

Mettre en place les services DHCP et DHCPv6 afin d'attribuer automatiquement les addresses IP aux hôtes du réseau, en utilisant un serveur, soit un routeur configuré comme serveur DHCP.

1. Reproduisez cette topologie en configurant les services DHCP afin d'attribuer automatiquement les adresses IP aux dispositifs du réseau.

1-1 J'ai reproduit la topologie demandée.



1-2 Configuration du Serveur DHCP

Étapes de configuration

1.2.1 configuration du routeur pour IPv4

Router> enable

Router# configure terminal

Router(config)# interface gigabitethernet 0/0

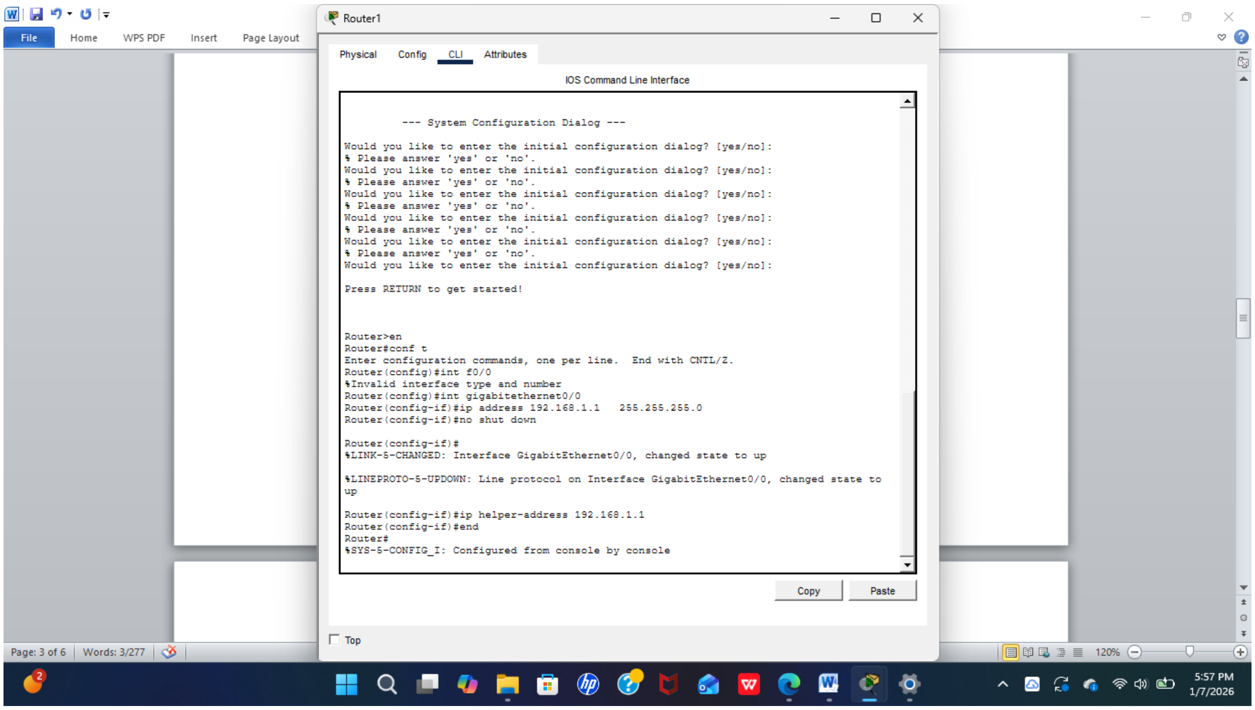
Router(config-if)# ip address 192.168.1.1 255.255.255.0

Router(config-if)# no shutdown

! Indiquer au routeur vers quel serveur DHCP renvoyer les requêtes

Router(config)# ip helper-address 192.168.1.1

Router(config)# end



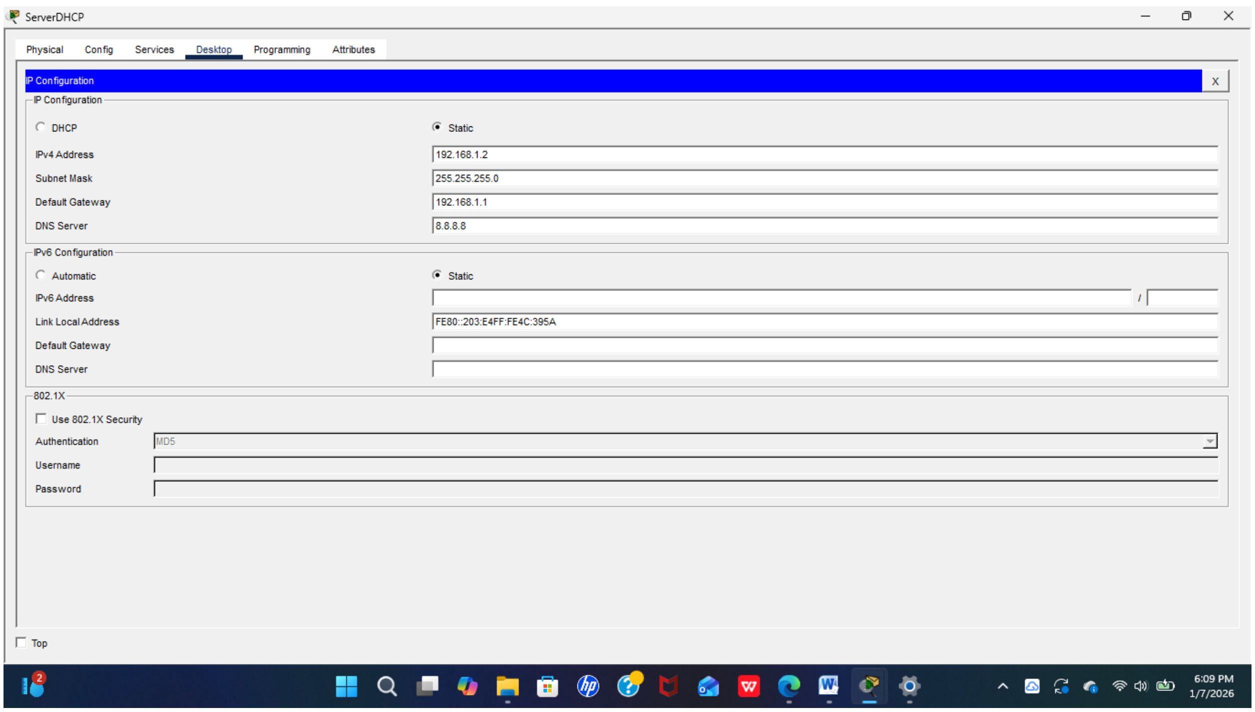
1.2.2 configuration du Serveur DHCP

Onglet Desktop →

IP Configuration : IP : 192.168.1.2

Mask : 255.255.255.0

Gateway : 192.168.1.1



Onglet Services → DHCP : Activer le service DHCP

Créer Pool1 (LAN1) :

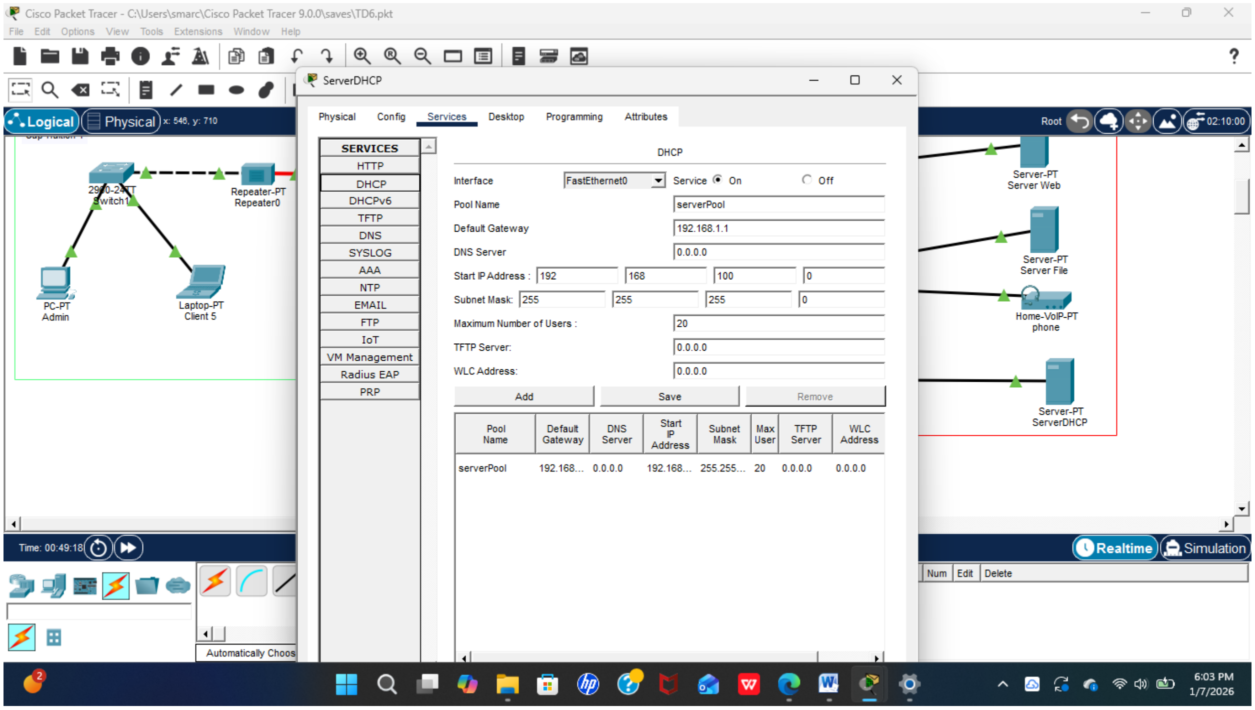
Default Gateway : 192.168.1.1

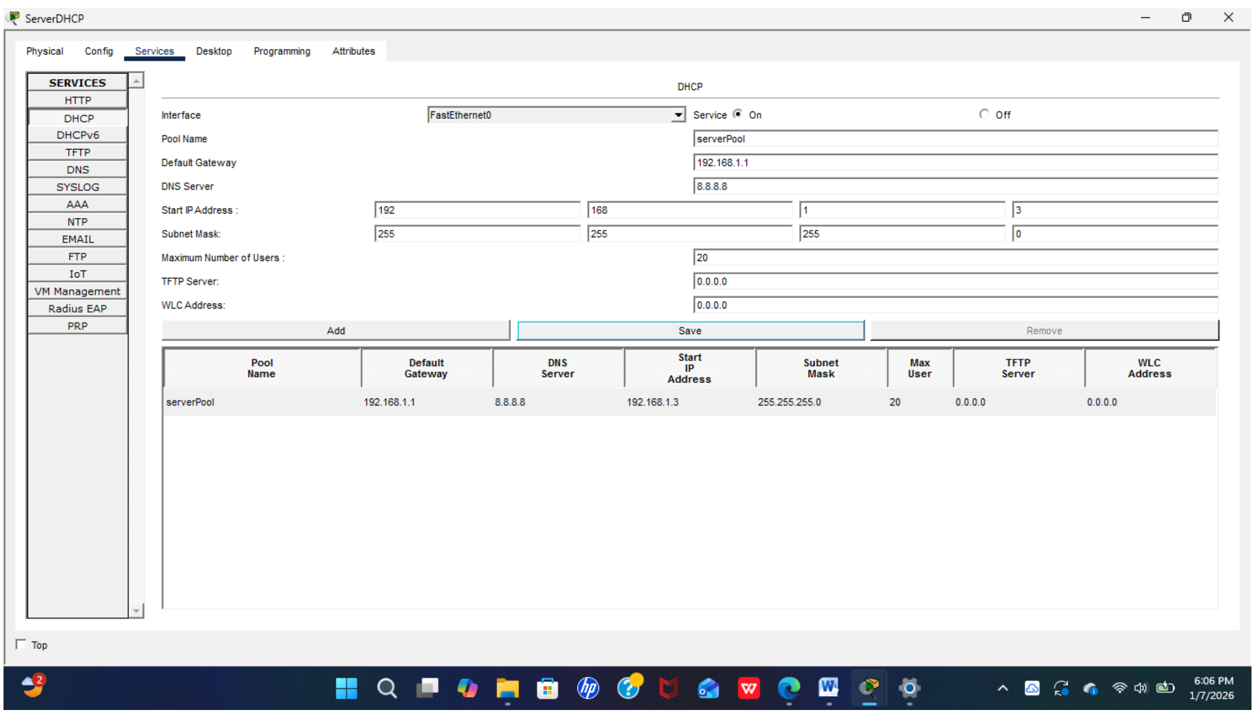
DNS Server : 8.8.8.8

Start IP : 192.168.1.3

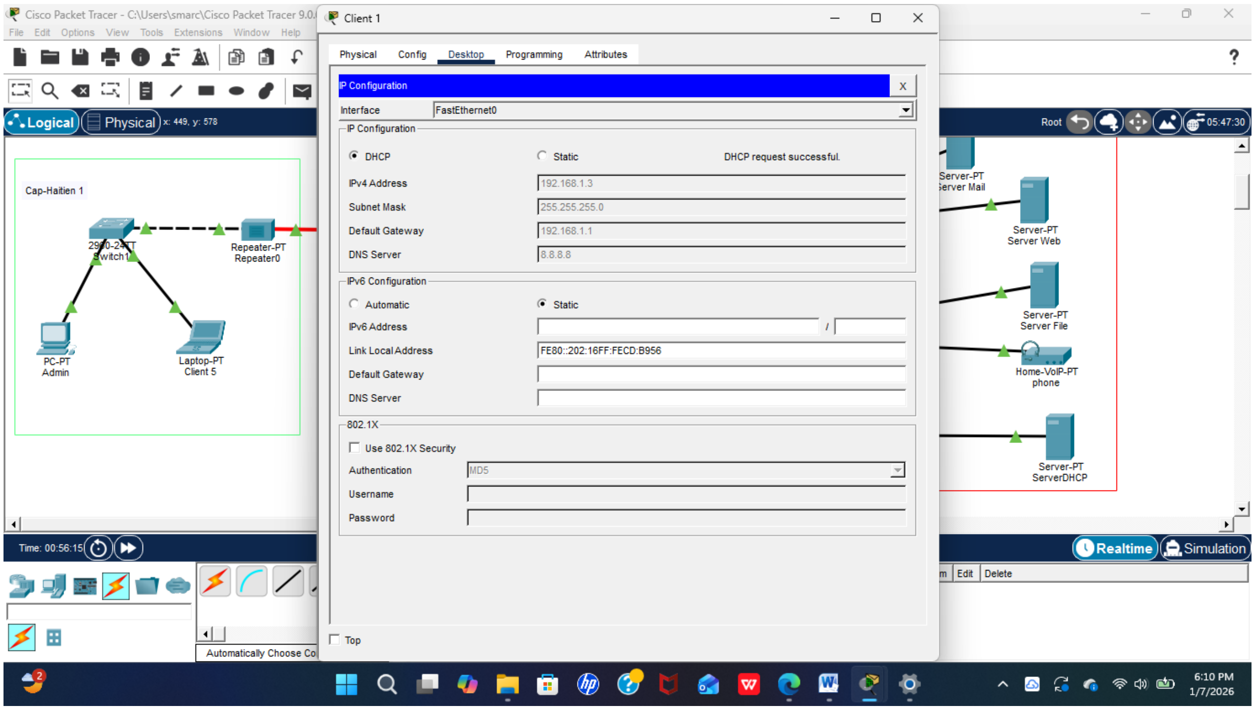
Subnet Mask : 255.255.255.0

Max Users : 20

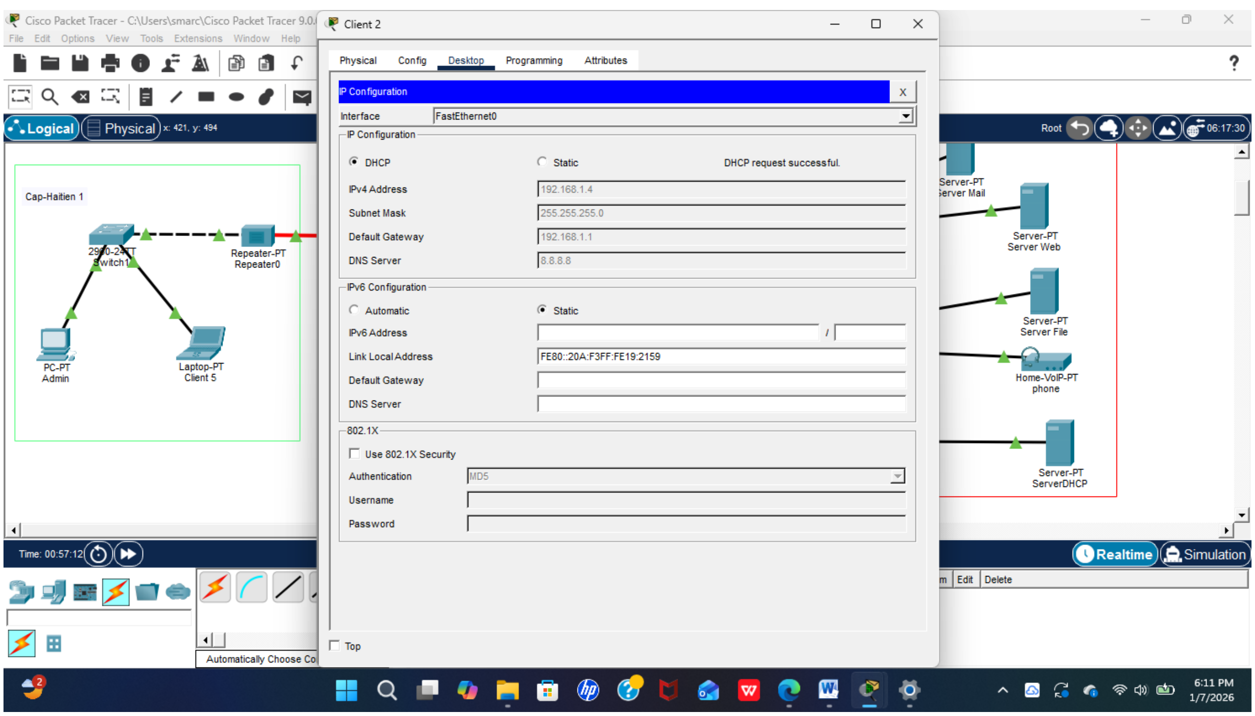

Cisco Packet Tracer - C:\Users\smarc\Cisco Packet Tracer 9.0.0\saves\TD6.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 546, y: 710
2960-24TT
Switch1
Repeater-PT
Repeater0
PC-PT
Admin
Laptop-PT
Client 5
ServerDHCP
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
Start IP Address :
192
168
100
0
Subnet Mask:
255
255
255
0
Maximum Number of Users :
20
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
192.168...
0.0.0.0
192.168...
255.255...
20
0.0.0.0
0.0.0.0
Root
02:10:00
Server-PT
Server Web
Server-PT
Server File
Home-VoIP-PT
phone
Server-PT
ServerDHCP
Time: 00:49:18
Realtime
Simulation
Num
Edit
Delete
Automatically Choos
6:03 PM
1/7/2026


ServerDHCP
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
192.168.1.1
DNS Server
8.8.8.8
Start IP Address :
192
168
1
3
Subnet Mask:
255
255
255
0
Maximum Number of Users :
20
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
192.168.1.1
8.8.8.8
192.168.1.3
255.255.255.0
20
0.0.0.0
0.0.0.0
Top
6:06 PM
1/7/2026

Exemple de DHCP4 sur client 1

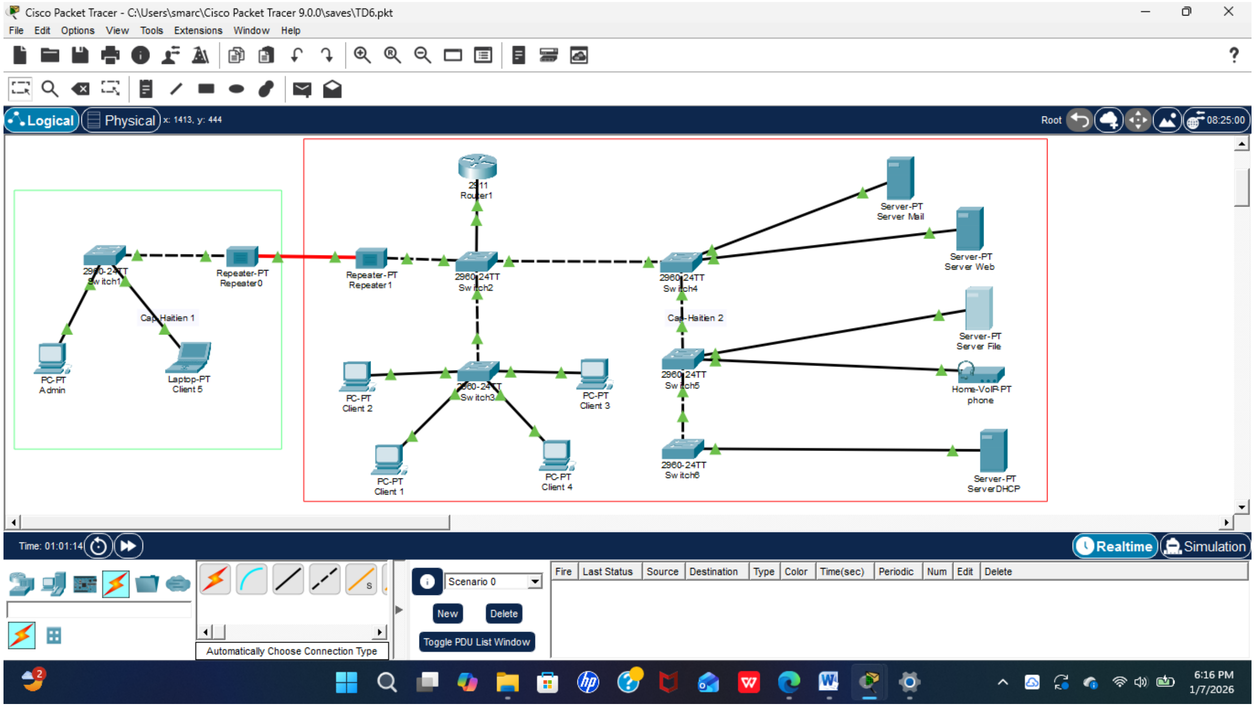


Sur client 2



2- Reproduisez cette topologie en configurant les services DHCPv6 afin d'attribuer automatiquement les adresses IP aux dispositifs du réseau.

2.1 la topologie reproduite



2.2 Configuration du Serveur DHCPv6

2.2.1 configuration du routeur

R1> enable

R1# configure terminal

R1(config)# ipv6 unicast-routing

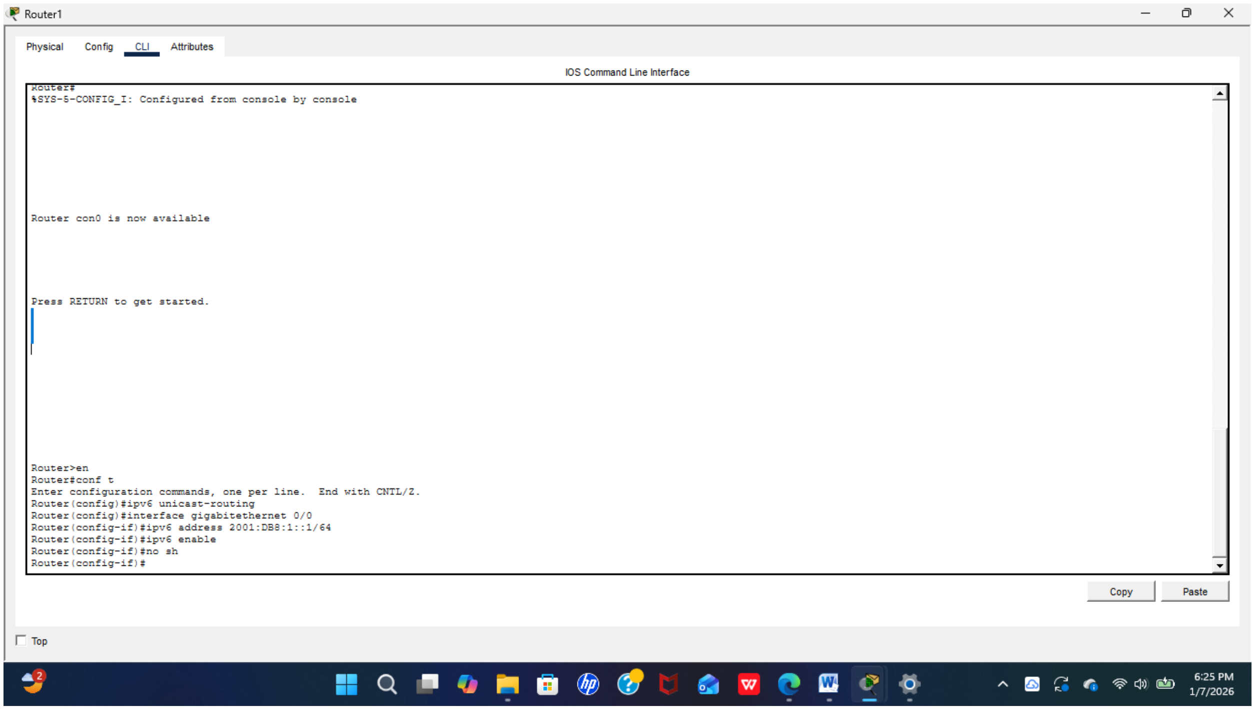
! Interface Cap 1

R1(config)# interface gigabitethernet 0/0

R1(config-if)# ipv6 address 2001:DB8:1::1/64

R1(config-if)# ipv6 enable

R1(config-if)# no shutdown



2.2.2 Configuration du Serveur DHCPv6

Aller dans Desktop → IP Configuration :

IPv6 Address : 2001:DB8:1::2

Prefix : /64

Gateway : 2001:DB8:1::1

Aller dans Services → DHCPv6 :

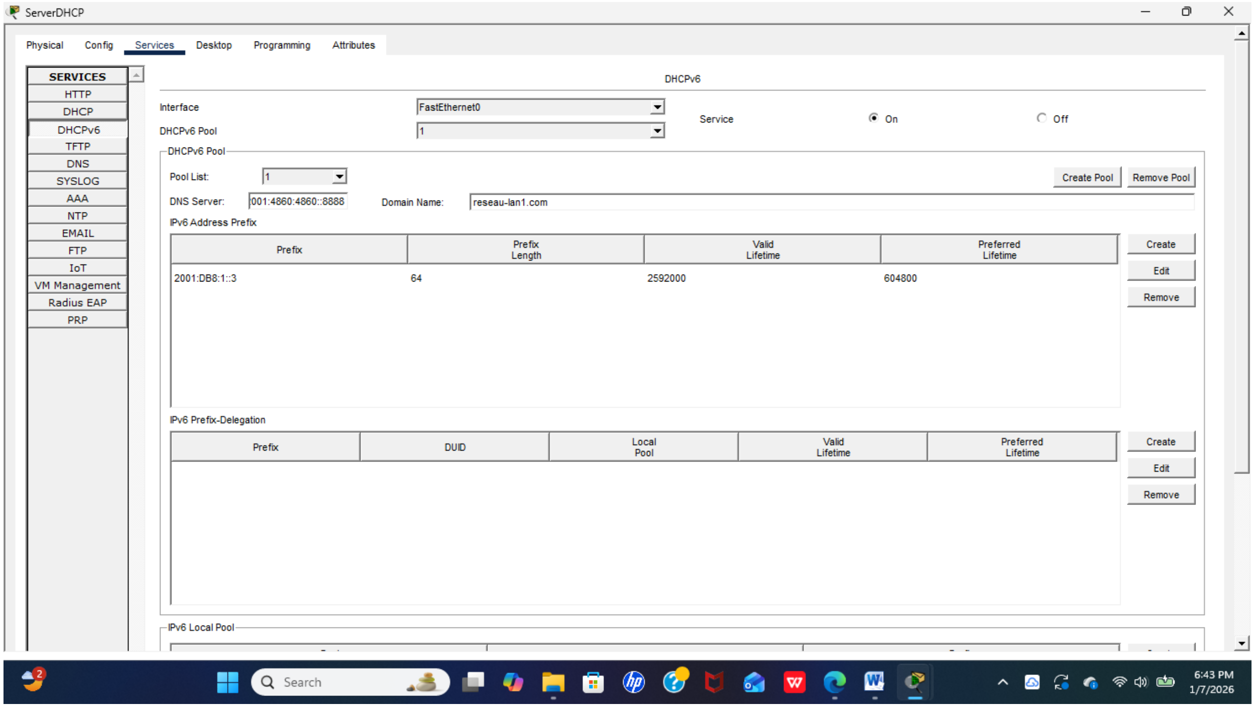
Activer le service DHCPv6 (On)

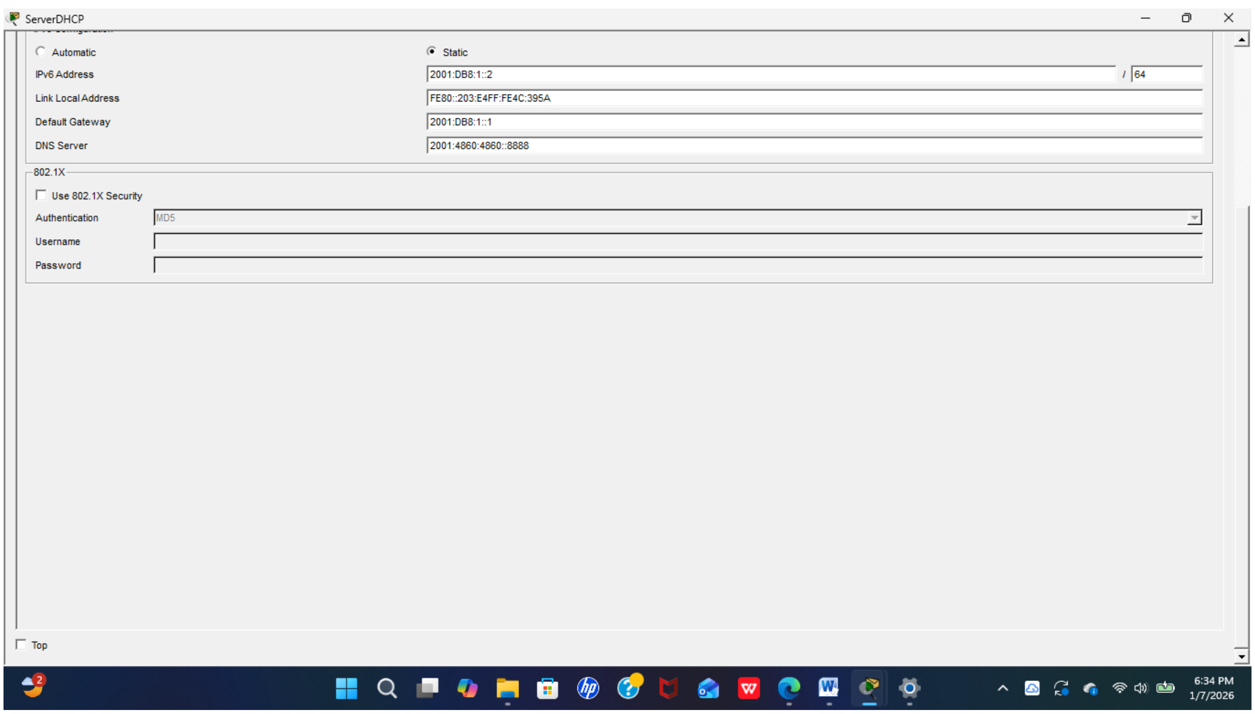
Ajouter Pool1 :

Prefix : 2001:DB8:1::/64

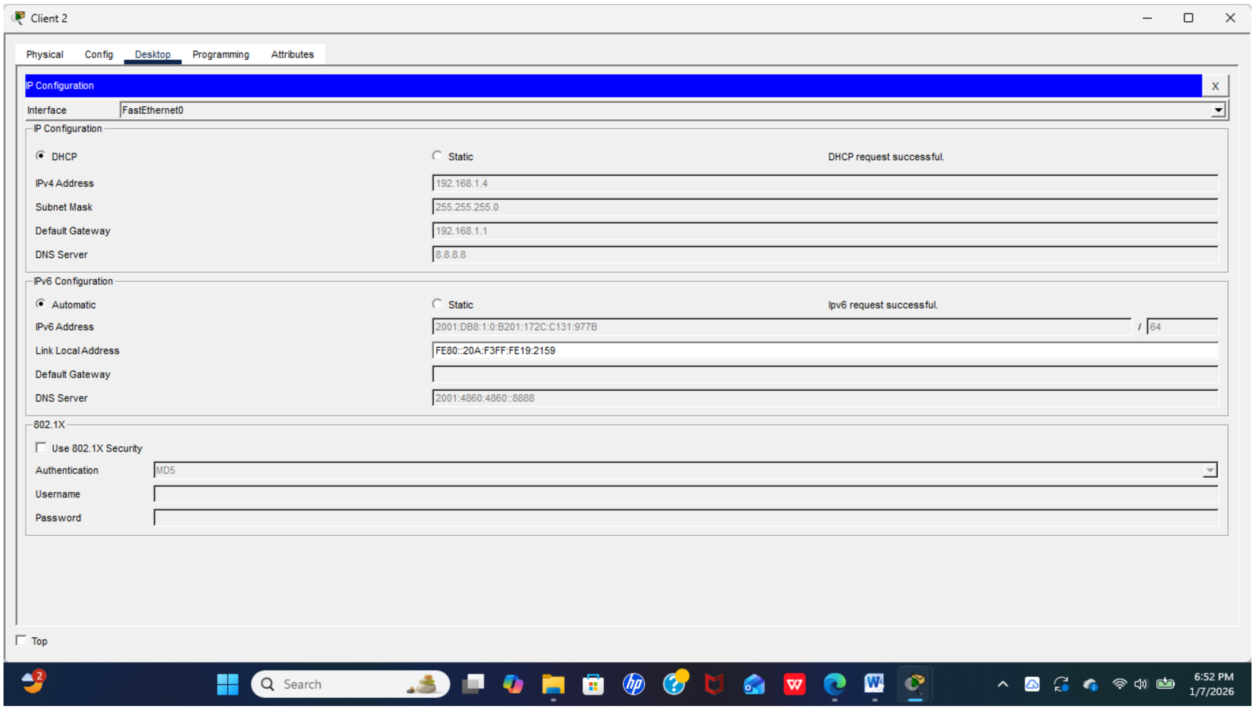
DNS Server : 2001:4860:4860::8888

Domain Name : reseau-lan1.com

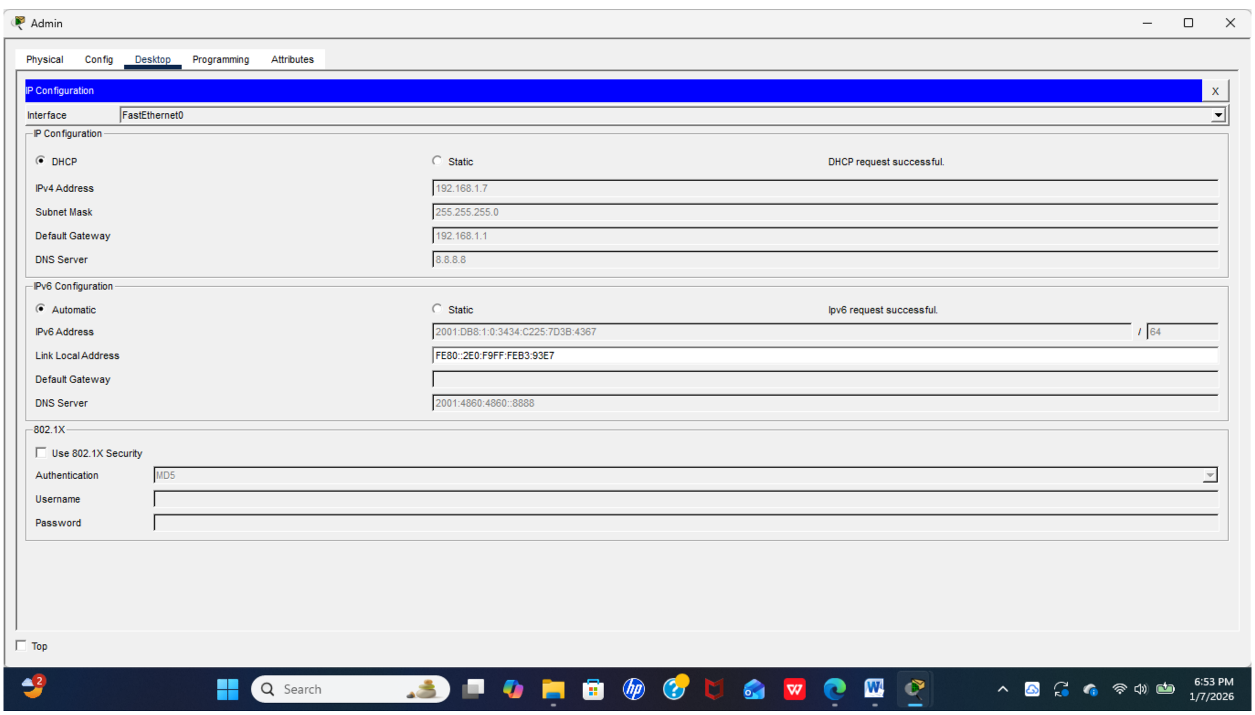




Exemple DHCPv6 sur client 2

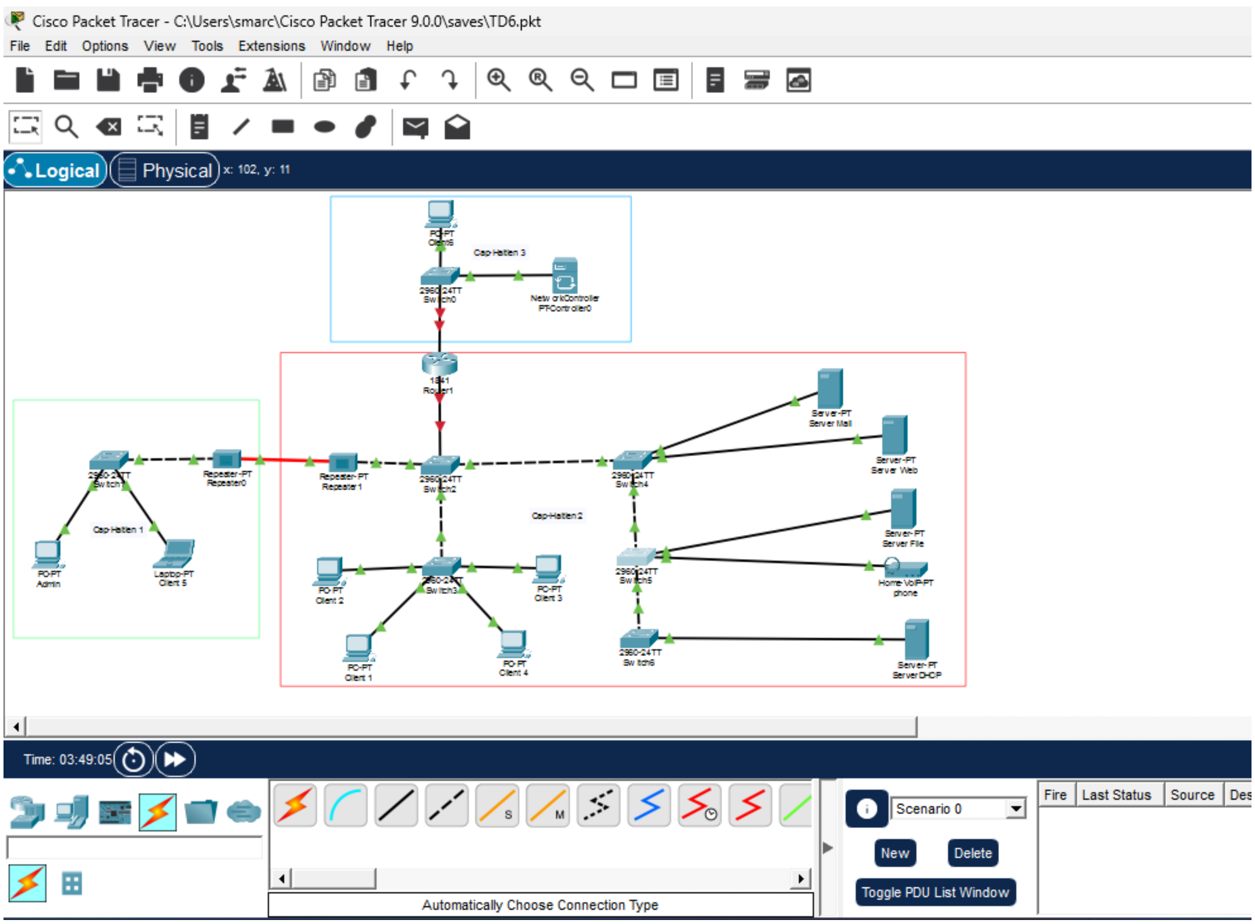


Sur admin



3. Reproduisez cette topologie en configurant les services DHCP, en utilisant le routeur comme serveur DHCP afin d'attribuer automatiquement les adresses IP aux différents hôtes du réseau.

3.1 Création de la topologie



3.2 configuration du service DHCP Sur le routeur

3.2.1 configuration des interface du routeur

R1> enable

R1# configure terminal

R1(config)# hostname R1

R1 ! Interface Cap1

R1(config)# interface gigabitEthernet 0/0

R1(config-if)# ip address 192.168.1.1   255.255.255.0

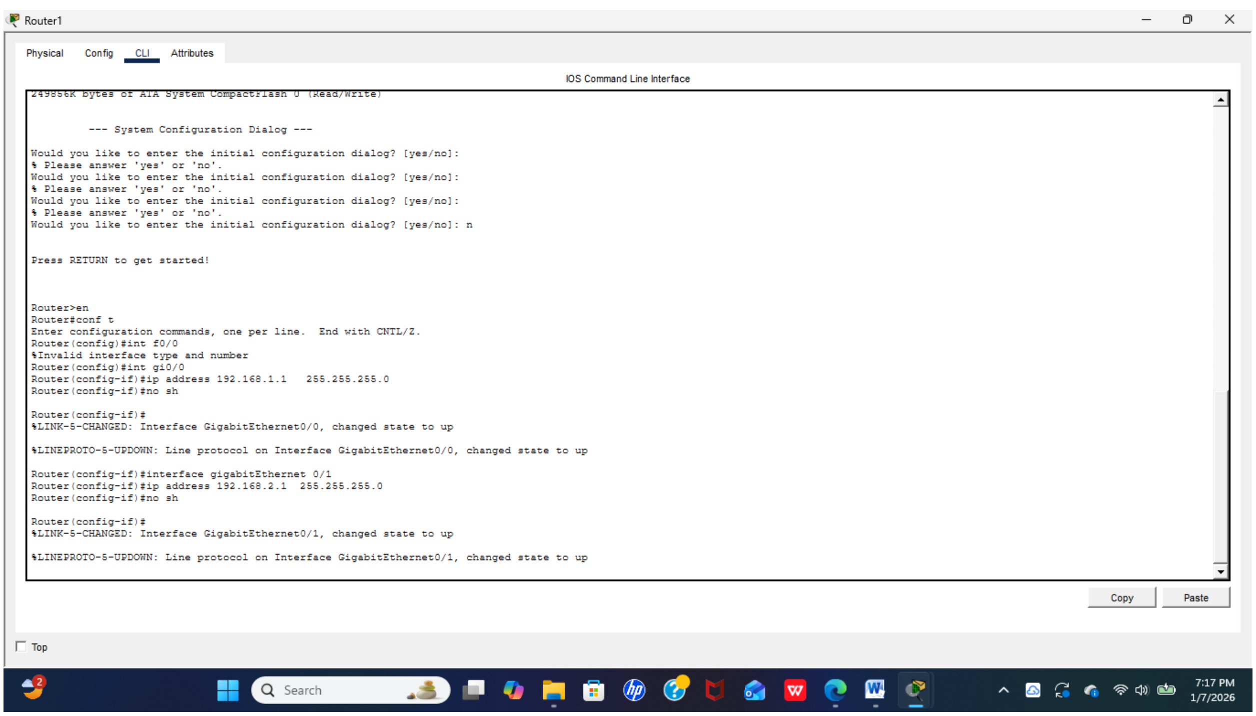
R1(config-if)# no shutdown

Interface Cap2

R1(config)# interface gigabitEthernet 0/1

R1(config-if)# ip address 192.168.2.1 255.255.255.0

R1(config-if)# no shutdown



3.2.2 Configuration du service DHCP sur le routeur

Pool DHCP pour CAP 1

R1(config)# ip dhcp pool Cap1

R1(dhcp-config)# network 192.168.1.0 255.255.255.0

R1(dhcp-config)# default-router 192.168.1.1

R1(dhcp-config)# dns-server 8.8.8.8

R1(dhcp-config)# exit

! Pool DHCP pour Cap2

R1(config)# ip dhcp pool Cap2

R1(dhcp-config)# network 192.168.2.0 255.255.255.0

R1(dhcp-config)# default-router 192.168.2.1

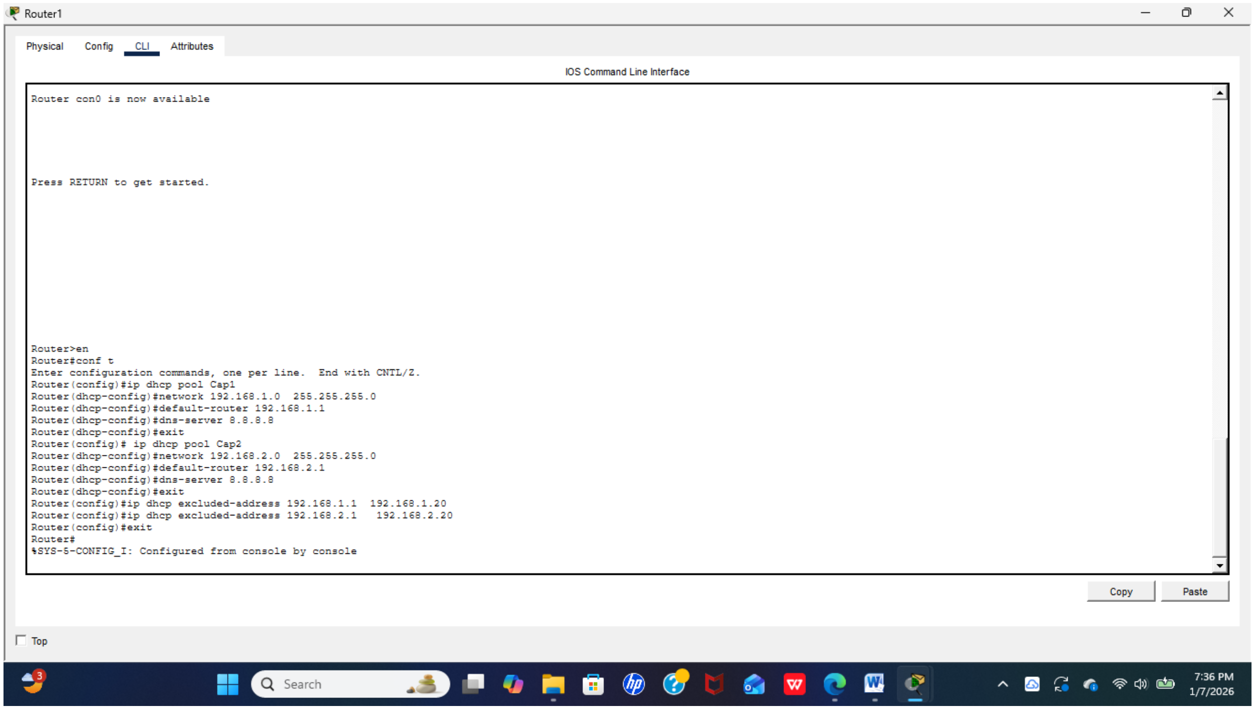
R1(dhcp-config)# dns-server 8.8.8.8

R1(dhcp-config)# exit

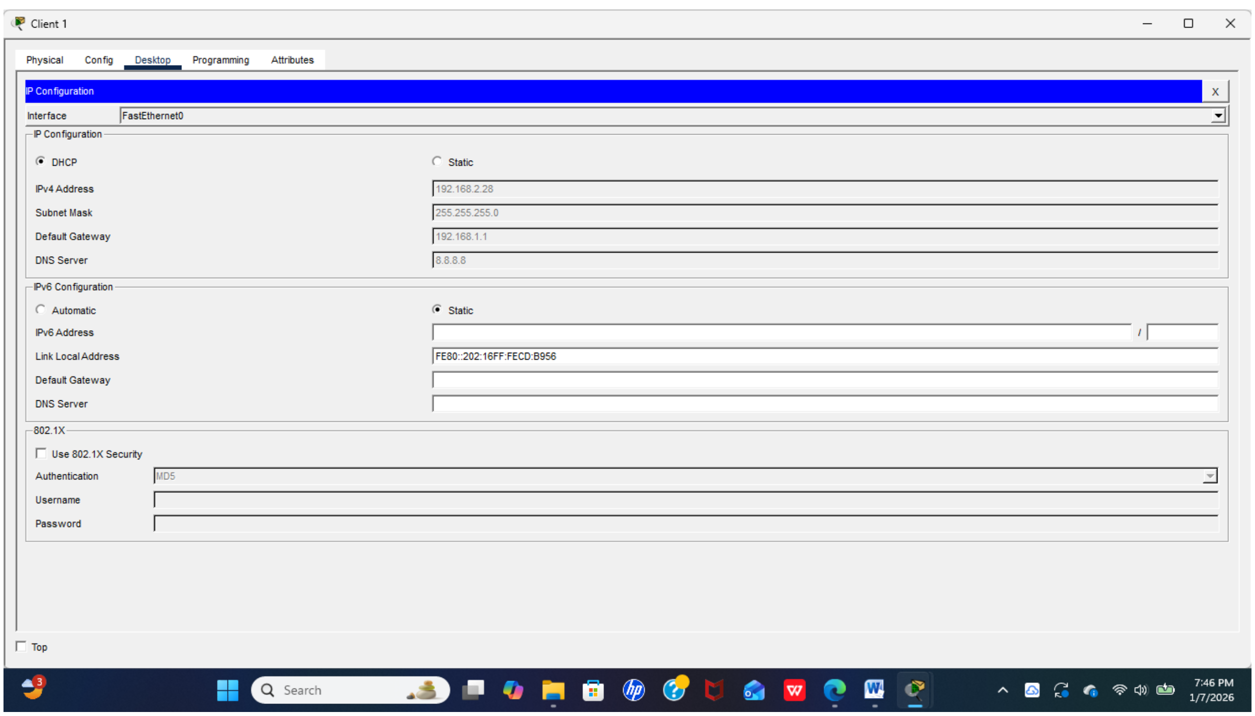
! Exclure les adresses réservées (passerelles, serveurs…)

R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.20

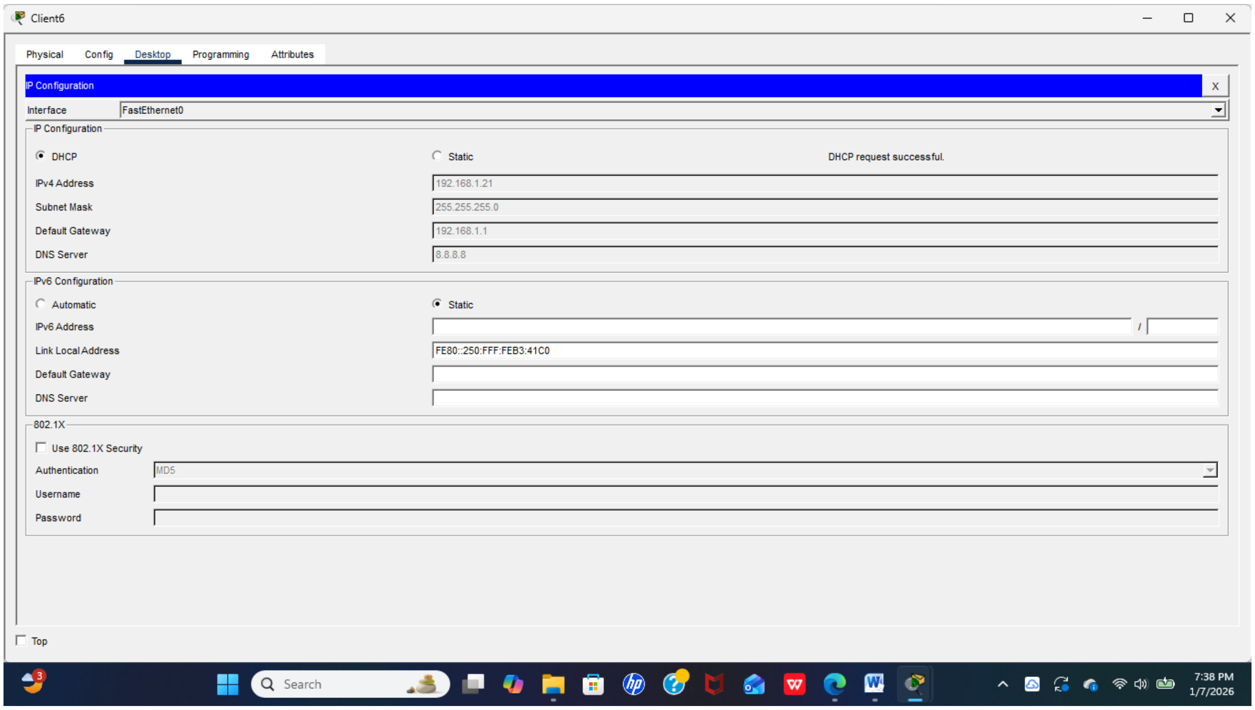
R1(config)# ip dhcp excluded-address 192.168.2.1   192.168.2.20



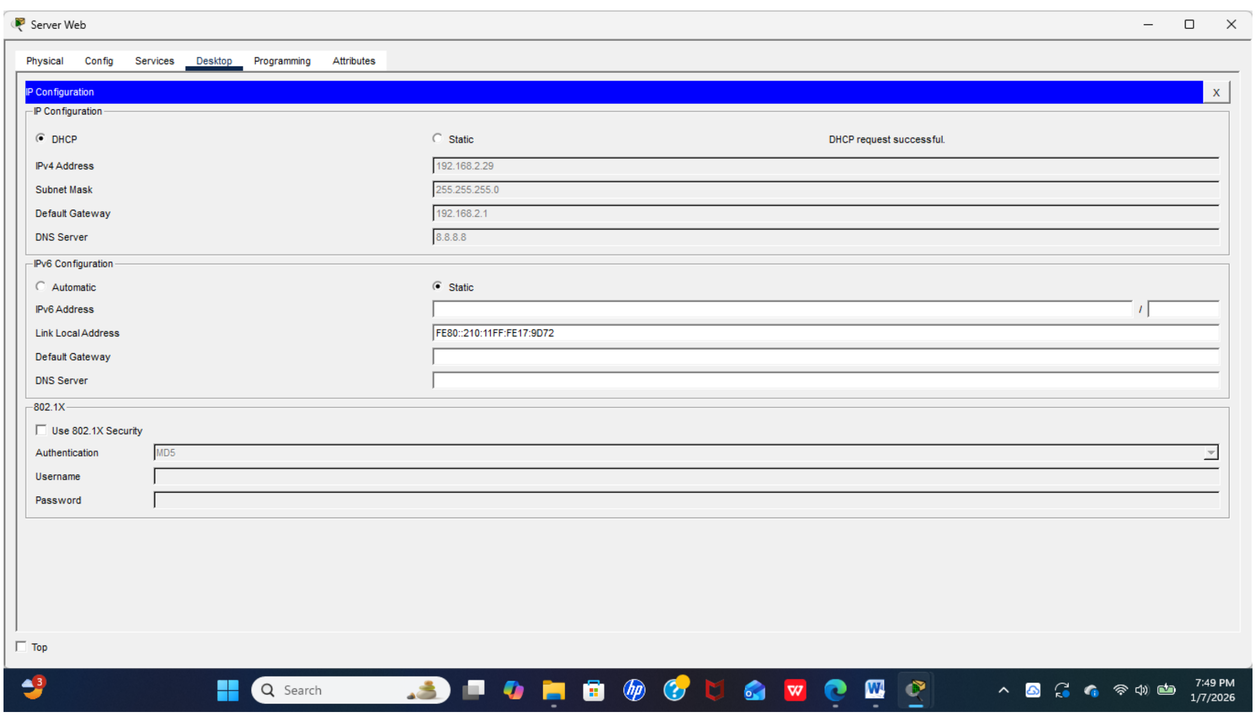
Exemple sur client 1 de Cap 2



Cap1, Client 6



Serveur web



3.2.3 Test de connectivité

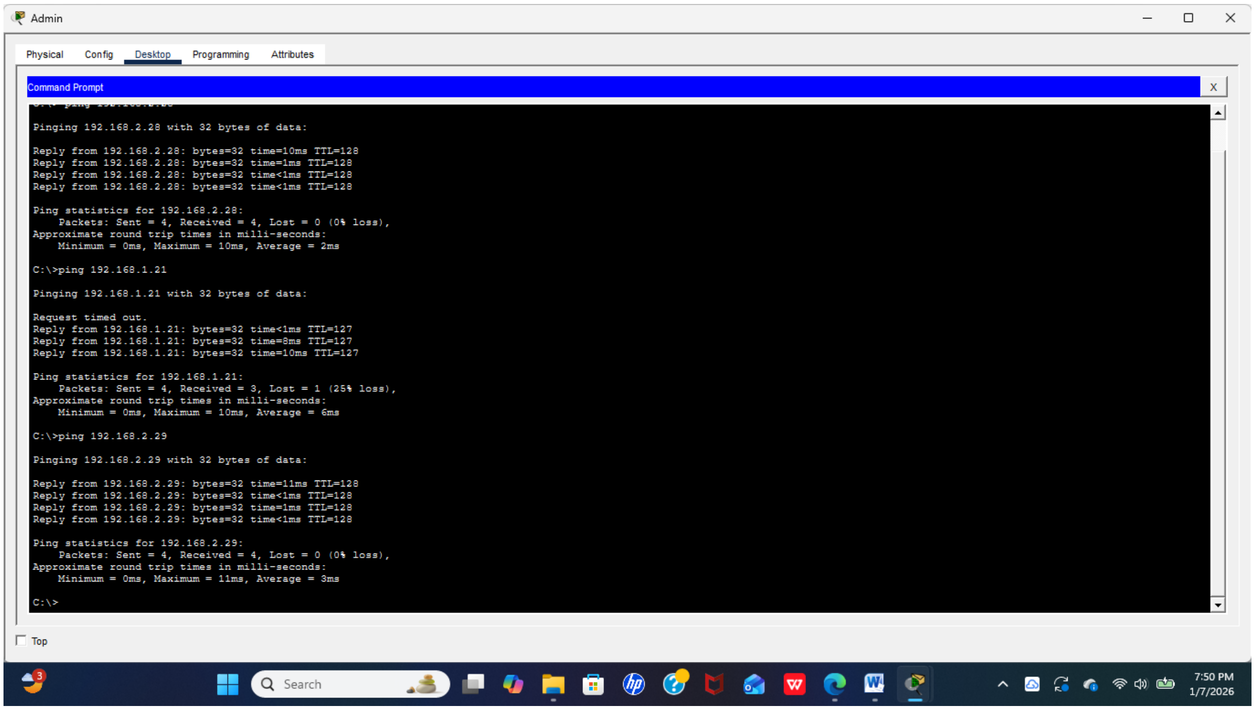
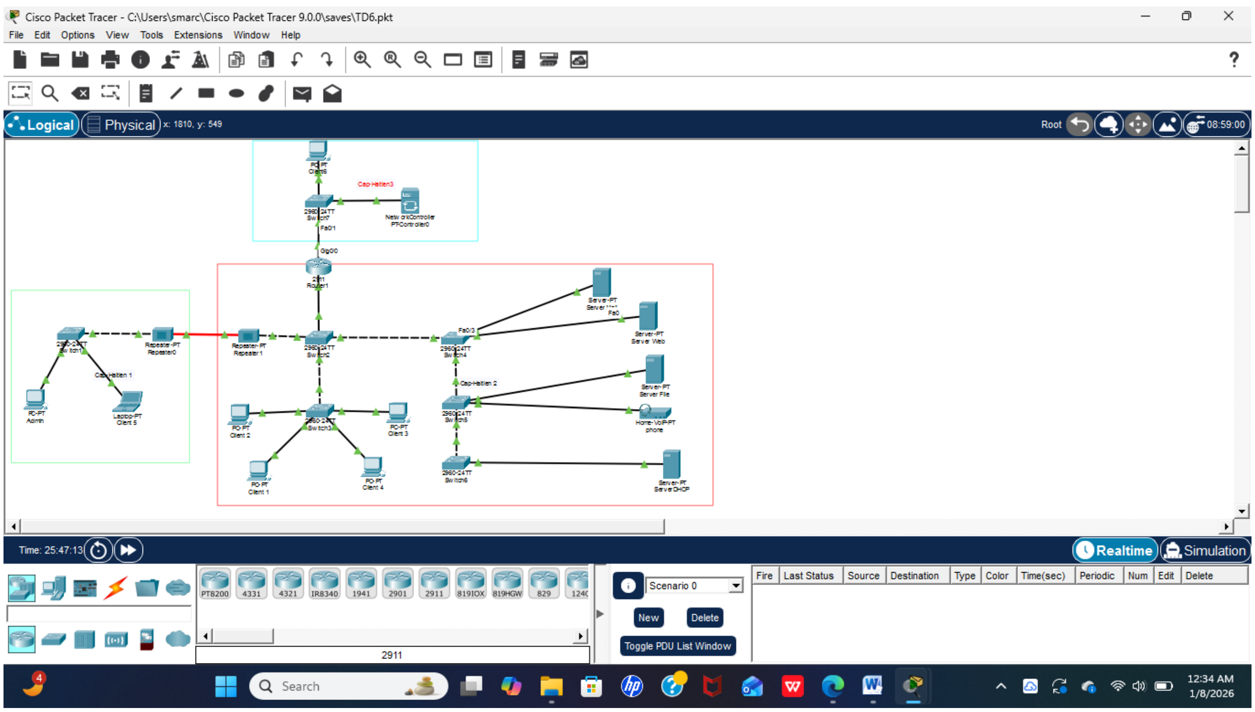


Figure 1 sur Admin j'ai epinglé 3 pcs , il y a connectivité entre les appareils.

4- Reproduisez cette topologie en configurant les services DHCPv6, en utilisant le routeur comme serveur DHCP afin d'attribuer automatiquement les adresses IP aux différents hôtes du réseau.

4.1 Topologie



4.2 Service DHCPv6

Étapes de configuration

Activer IPv6 sur le routeur

Configurer les interfaces

Créer les pools DHCPv6

R1> enable

R1# configure terminal

R1(config)# ipv6 unicast-routing

! Interface Cap1

R1(config)# interface gigabitEthernet 0/0

R1(config-if)# ipv6 address 2001:DB8:1::1/64

R1(config-if)# ipv6 enable

R1(config-if)# no shutdown

! Interface Cap2

R1(config)# interface gigabitEthernet 0/1

R1(config-if)# ipv6 address 2001:DB8:2::1/64

R1(config-if)# ipv6 enable

R1(config-if)# no shutdown

! Pool pour Cap1

R1(config)# ipv6 dhcp pool Cap1

R1(config-dhcpv6)# address prefix 2001:DB8:1::/64

R1(config-dhcpv6)# dns-server 2001:4860:4860::8888

R1(config-dhcpv6)# domain-name reseau-lan1.com

R1(config-dhcpv6)# exit

! Pool pour Cap2

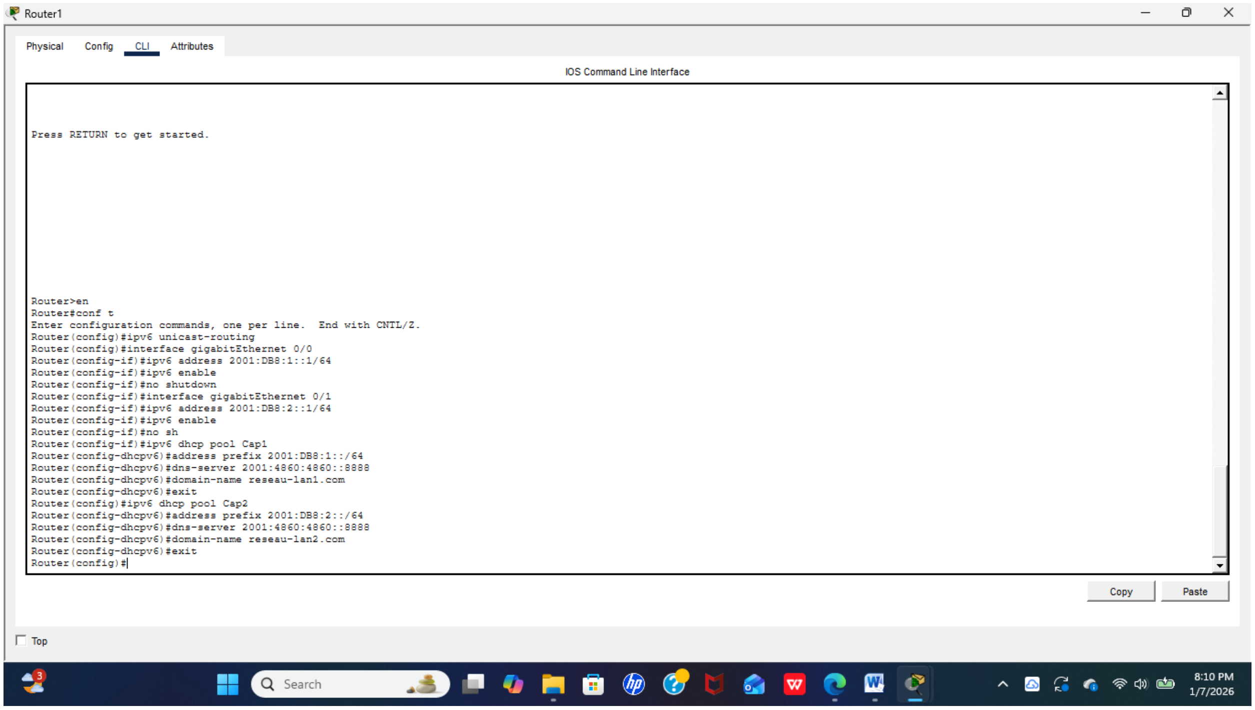
R1(config)# ipv6 dhcp pool Cap2

R1(config-dhcpv6)# address prefix 2001:DB8:2::/64

R1(config-dhcpv6)# dns-server 2001:4860:4860::8888

R1(config-dhcpv6)# domain-name reseau-lan2.com

R1(config-dhcpv6)# exit



Associer DHCPv6 aux interfaces

! Interface Cap1

R1(config)# interface gigabitEthernet 0/0

R1(config-if)# ipv6 dhcp server Cap1

R1(config-if)# ipv6 nd managed-config-flag

R1(config-if)# exit

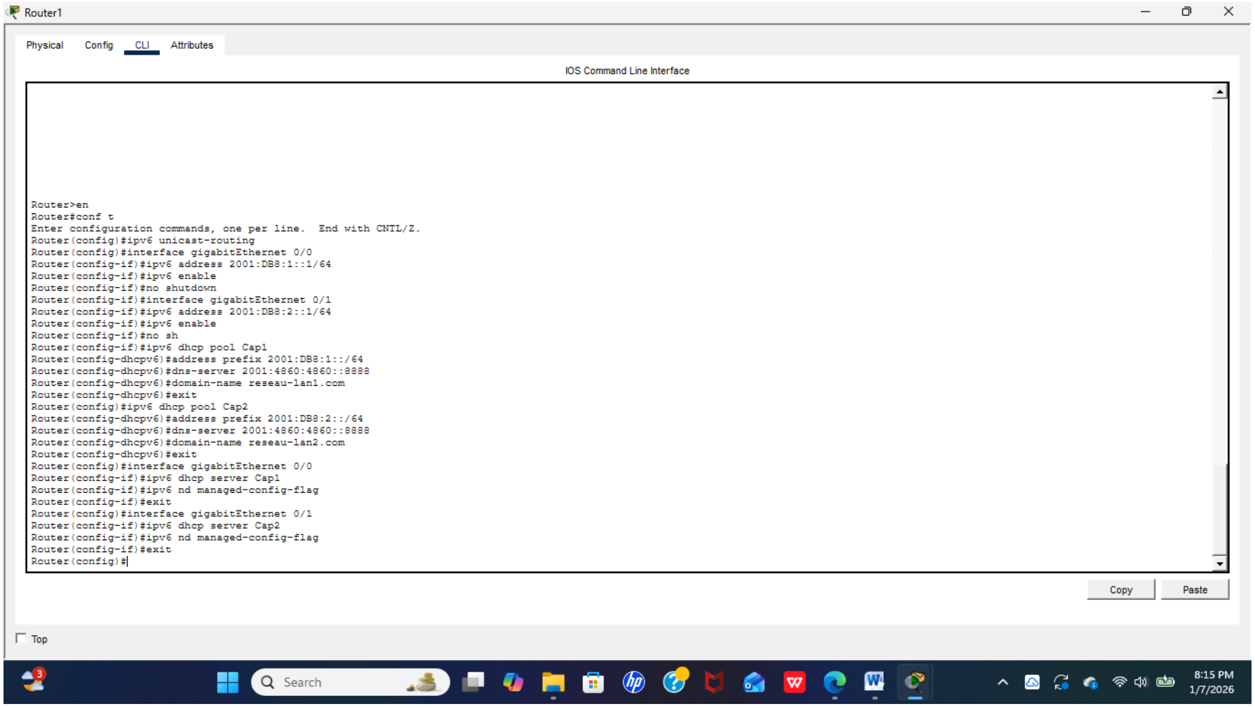
! Interface Cap2

R1(config)# interface gigabitEthernet 0/1

R1(config-if)# ipv6 dhcp server Cap2

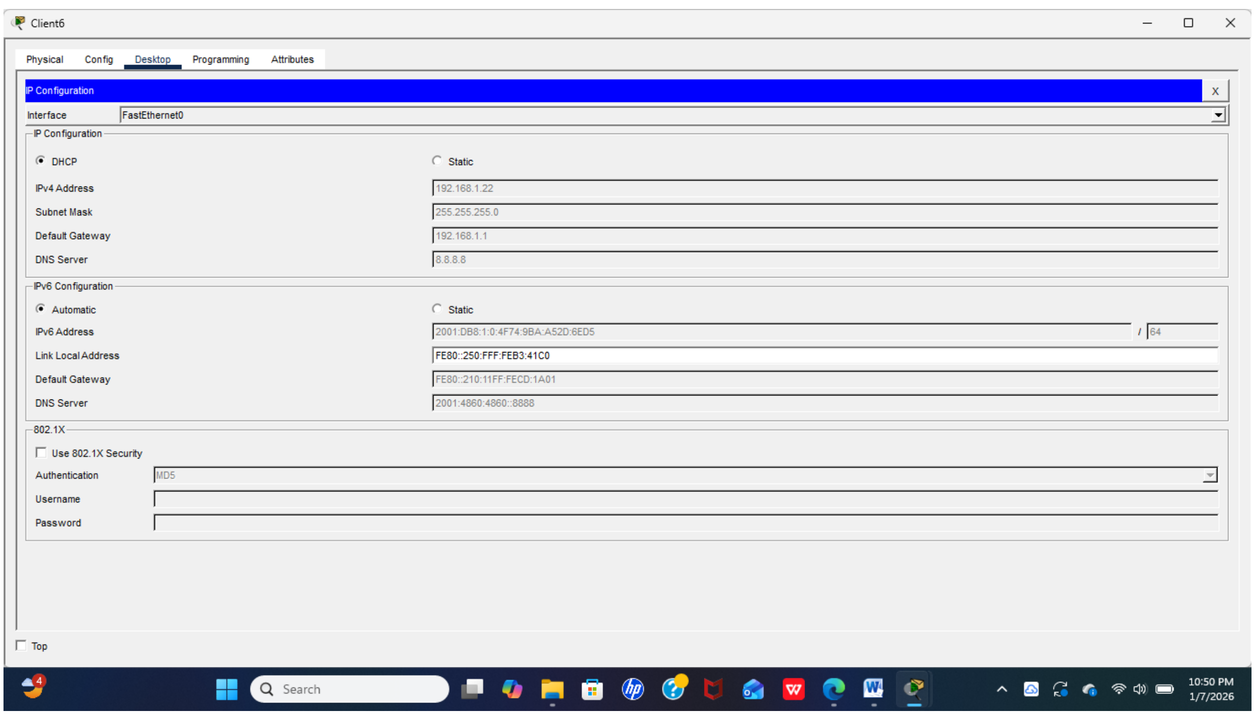
R1(config-if)# ipv6 nd managed-config-flag

R1(config-if)# exit

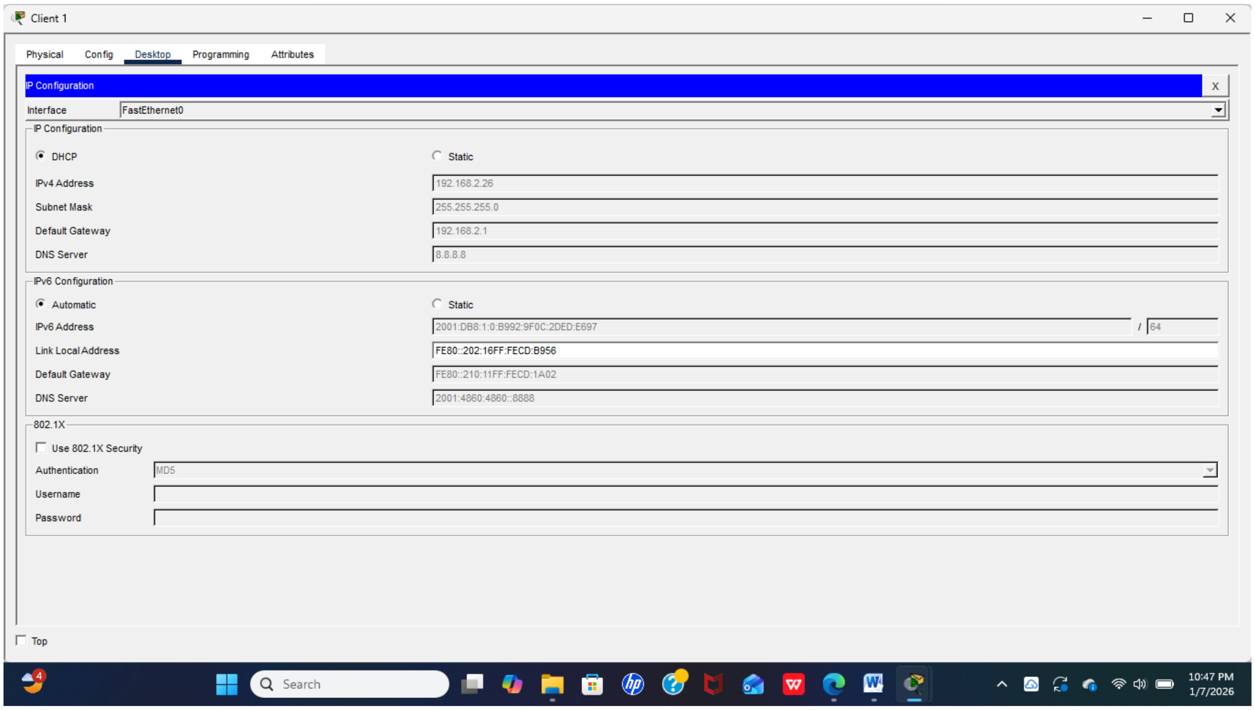


4.2.1 Configuration des appareil de manière automatique

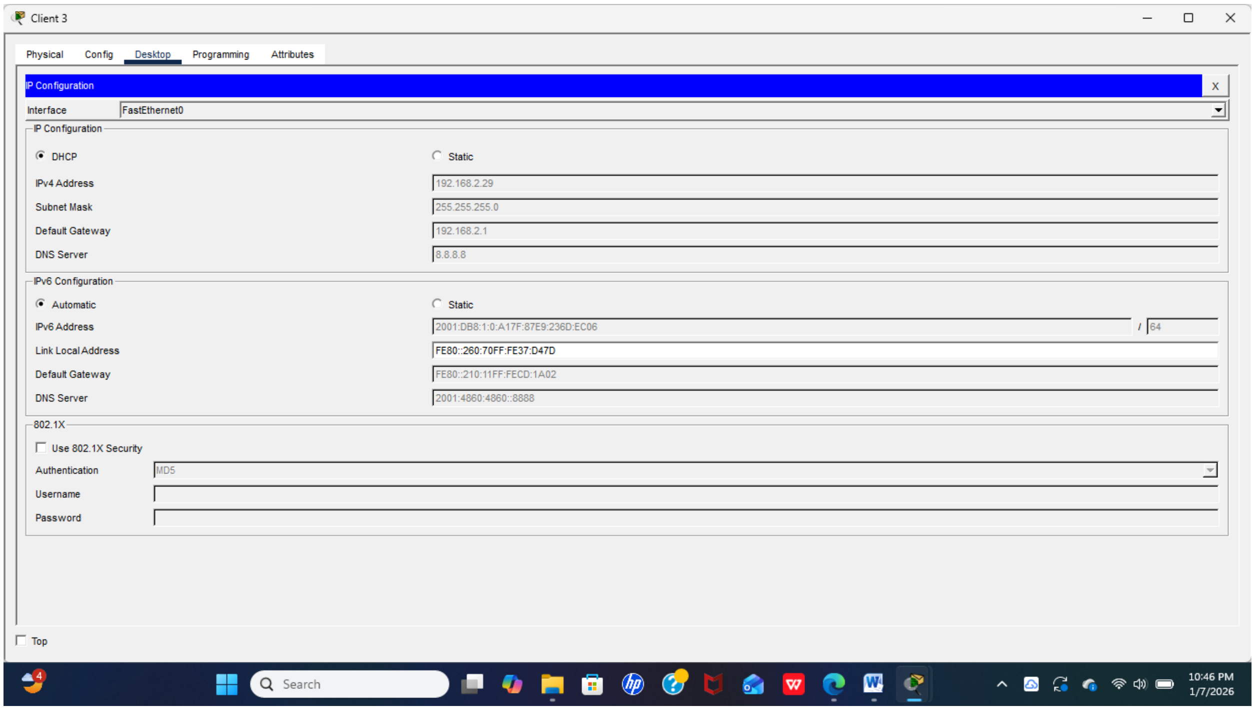
Client 6



Client 1


Client 1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.26
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1:0:B992:9F0C:2DED:E697
64
Link Local Address
FE80::202:16FF:FECD:B956
Default Gateway
FE80::210:11FF:FECD:1A02
DNS Server
2001:4860:4860::8888
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Search
10:47 PM
1/7/2026

Client3


Client 3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.29
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1:0:A17F:87E9:236D:EC06
/ 64
Link Local Address
FE80::260:70FF:FE37:D47D
Default Gateway
FE80::210:11FF:FECD:1A02
DNS Server
2001:4860:4860::8888
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Search
10:46 PM
1/7/2026

4.2.2 Test de connectivité

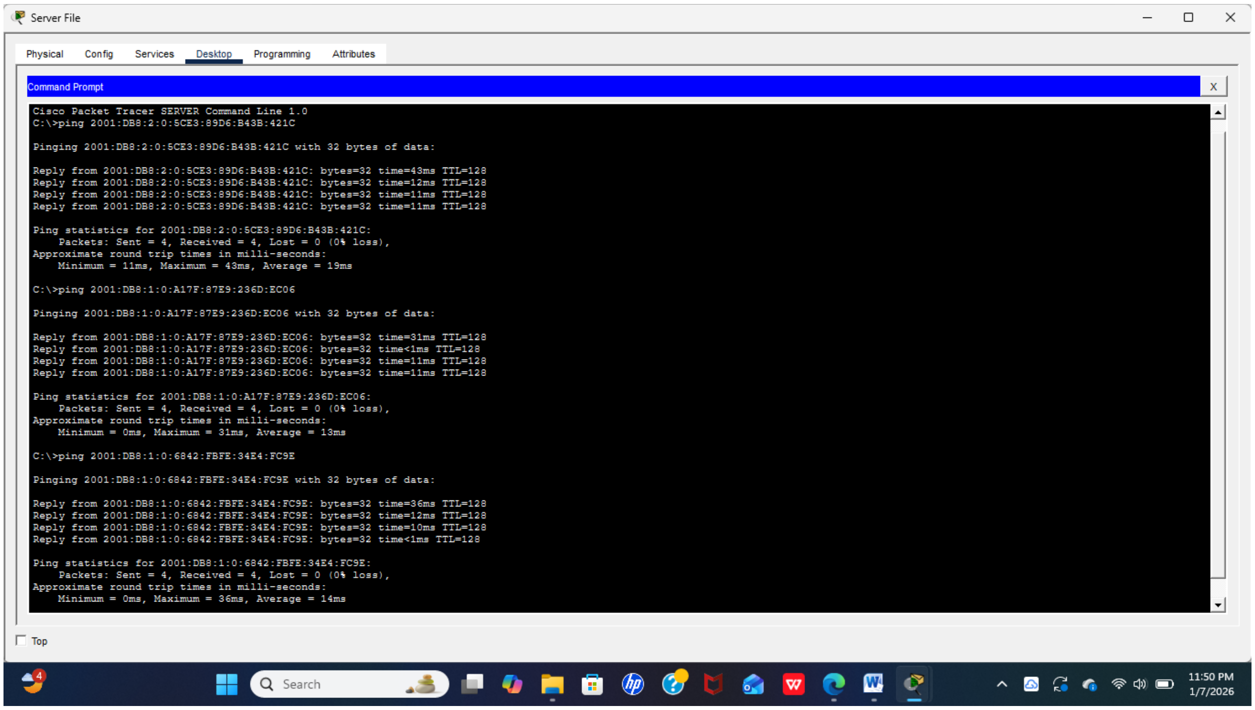


Figure 2 Sur le Serveur File j'ai epinglé 3 pcs, il y a connectivité entre les appareils.

Conclusion

En conclusion, ce TD a permis de configurer et de tester avec succès les services DHCP et DHCPv6. Les hôtes ont obtenu automatiquement leurs addresses IP, confirmant le bon fonctionnement du réseau et l'importance de l'automatisation de l'adressage.